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CircuitFlow: Smart Water Management, Drop by Drop

Urban Development, Mobility & Smart Cities: 73

Phase - 2

Go, change the world

Team

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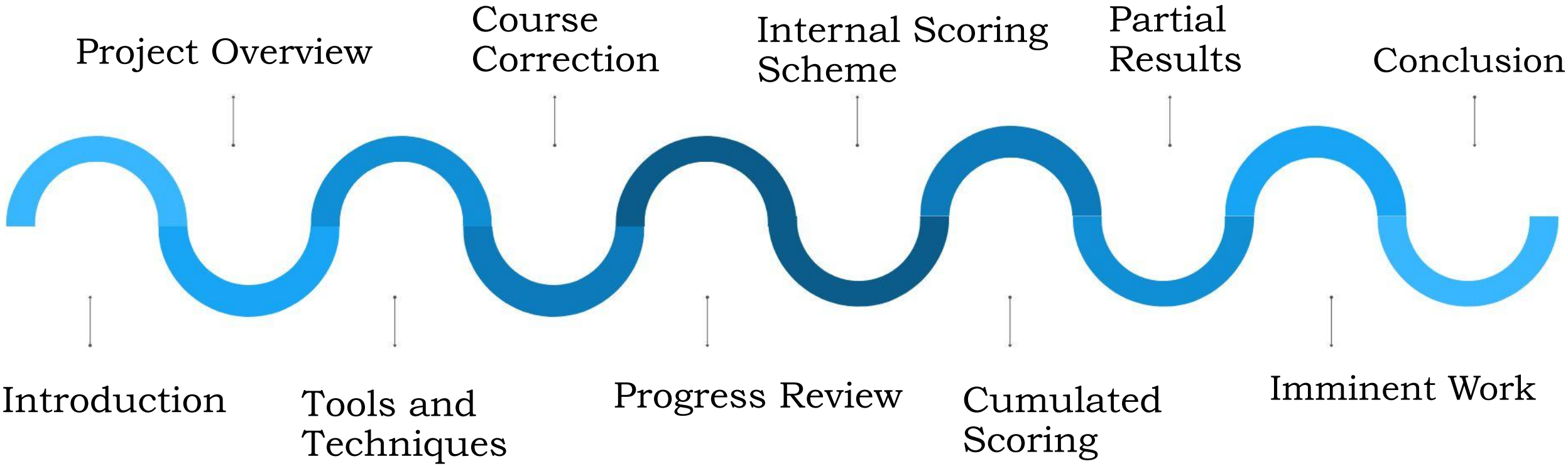


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Outline

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Introduction

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Overview of the topic

Smart tap-system for water wastage remediation and *usage analysis* at the household level through real-time data collection and *predictive modelling*.

- This is important because it leverages technology to create a more efficient and cost-effective system to address the critical global challenges of water scarcity.

- This model creates a structured wastage analysis and remediation system based on leakage-alerting, usage-dashboards, and data-driven regulation.

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Project Overview

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Problem Statement

- Rapid population growth and poor monitoring have intensified domestic water wastage, with Indian households losing about 45,500 L annually.
- Existing systems lack real-time control and user engagement.
- This project develops a smart, sensor-based system with flow-control valves and analytics to detect anomalies, automate corrections, and promote sustainable water use.

Significance

- Rapid population growth and urbanization are straining freshwater resources, while unmonitored household use and unnoticed leaks worsen wastage.
- Smart real-time monitoring can improve awareness, reduce losses, and promote sustainable water consumption in homes.

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Project Overview

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Problem Context
• Water scarcity has become a critical global issue, with studies indicating that throughout 2025 nearly 67% of the world's population will experience moderate to high water stress. • In India, urban households consume between 183 and 215 liters of water per person daily, placing immense pressure on already limited resources. • The residential sector accounts for the largest portion of urban water usage, primarily for bathing, washing, flushing, and cooking.Bath and shower accounts for 55% of water-usage in urban households.

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Core Objectives

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Main Objective
• To effectively identify and address unattended water loss in the households of major Indian cities.
Specific Objectives
• To develop a model that detects and remediates leaking and overflowing taps. • To demonstrate a dashboard on users' water utilization.
Expected Contributions
• The model proposes to add a data-driven, adaptive approach to existing models on the subject.

Course Correction

- **Scalability:** The concern on scalability was discovered and addressed through rigorous constraint analysis.
- **Hardware Refinement:** Multiple sensing approaches were narrowed down to a single consistent IR sensor.
- **Elimination of redundant hardware:** Redundant hardware components such as LCD display were dropped from the hardware architecture.

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Review of Novelty

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Overview

- Based on the literature review, the model was adapted with a proportional solenoid valve.

- The fundamental thesis revolves around:
  - **Local Governance**: Local processing and presentation of data.
  - **Data Sovereignty:** Security of user data.
  - User-friendly proportional valve.

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Objectives: Phase II

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- Accurately display *flow data* from the *water flow sensor* for fixed intervals of time, in order to understand variations in flow.

- To detect *hand proximity* using the *IR sensor*, enabling differentiation between active user interaction and unattended flow.

- Ensure proper *assembly and functioning* of core hardware components into a prototype

- To develop an *accessible user interface (UI)* that displays usage information in a visual and organised manner without requiring internet connectivity.

Tools and Techniques

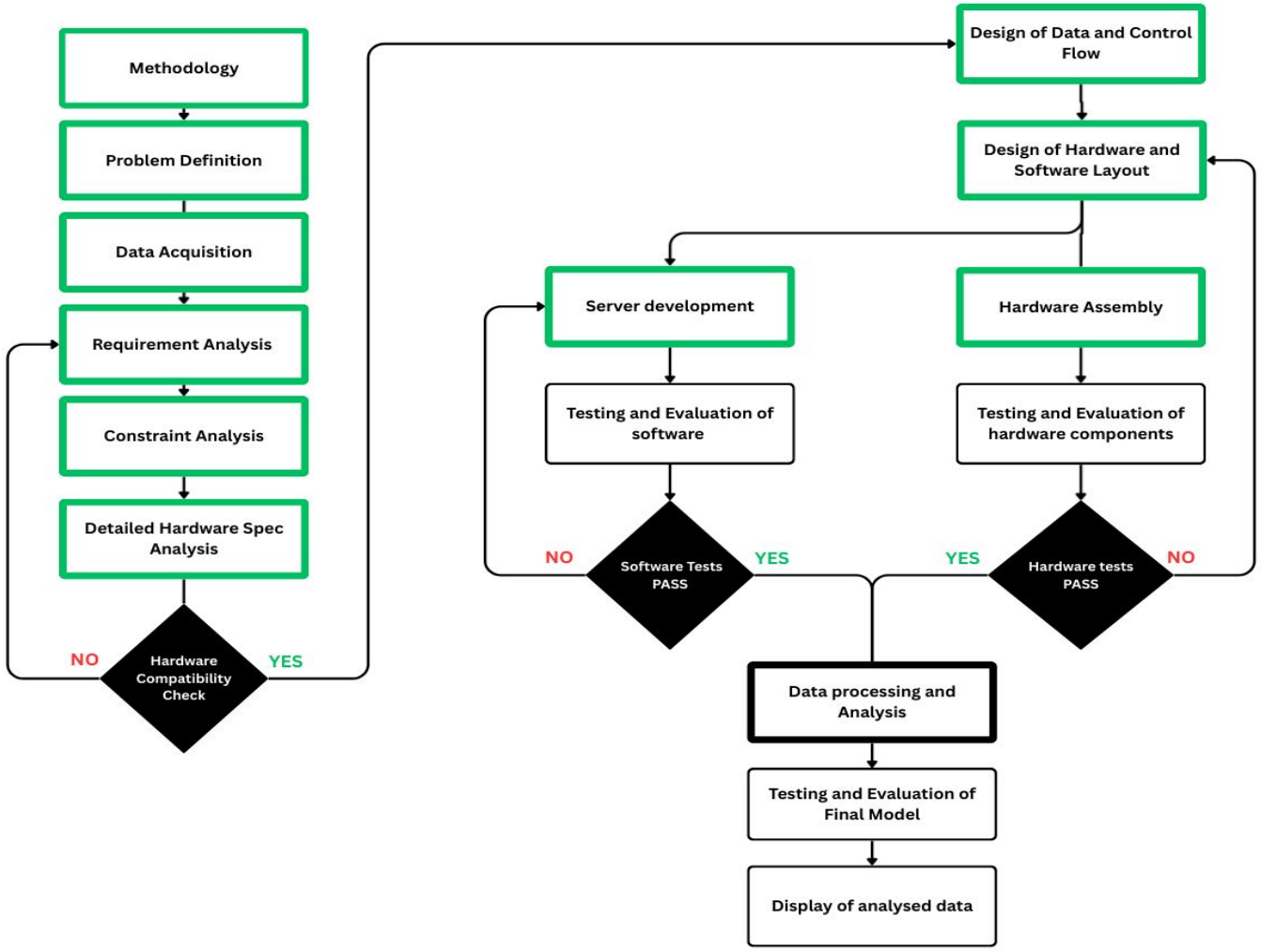
Hardware
Arduino UNO (Microcontroller Unit)
YF-S201 Water Flow Sensor
HW201 IR Obstacle Sensor
LEDs (Red & Green)
Buzzer
Jumper Wires, Resistors, Power Supply

Software
Flask
Pyserial
Multithreading
Web development tools (HTML, CSS, JavaScript)
Chart.js
Python

Progress Review

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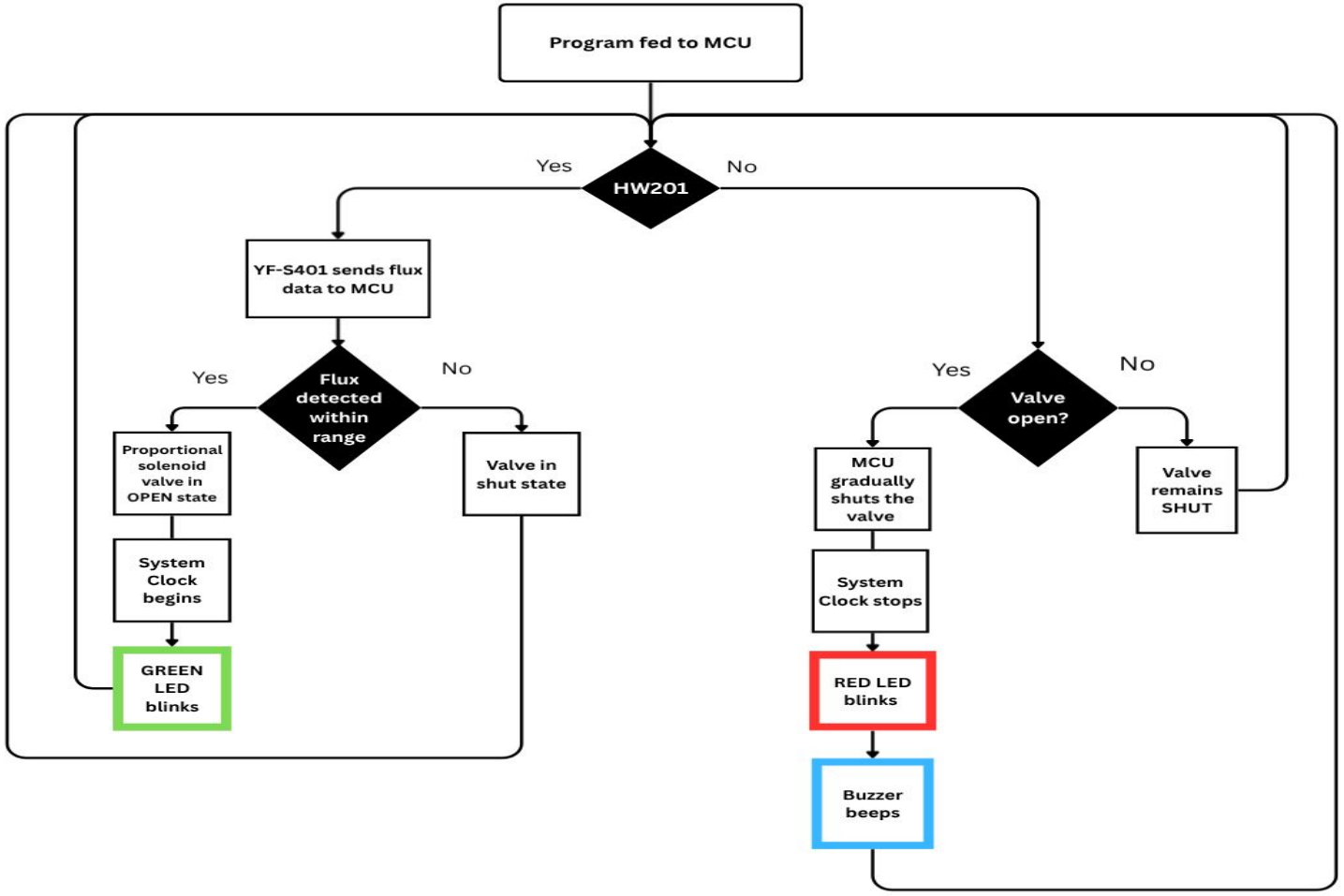
Plan of Action



Legend	
Methodology	Accomplished task
Testing and Evaluation of Final Model	Imminent Work
Data processing and Analysis	Critical Task

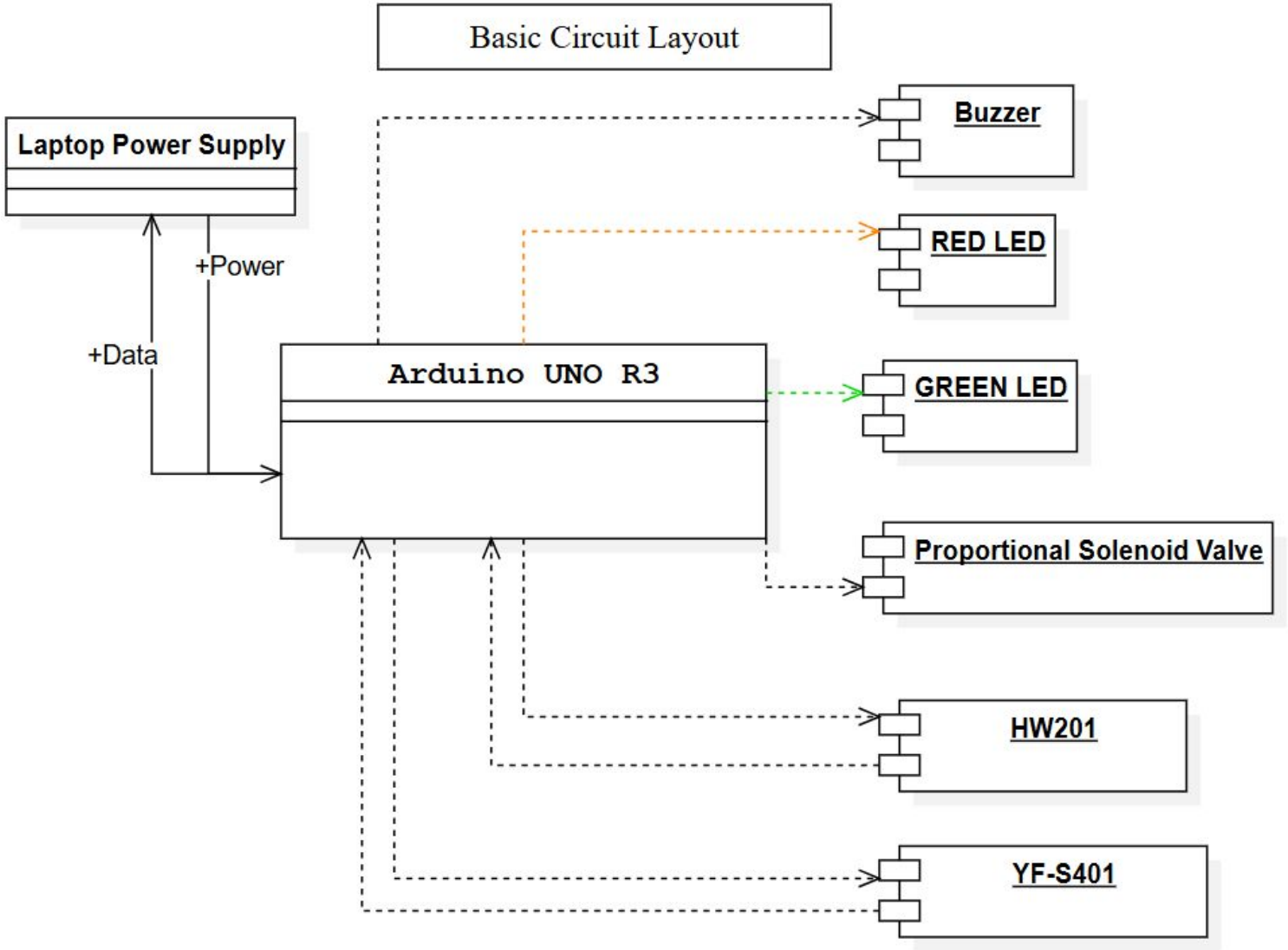
Progress Review

System Architecture



Progress Review

Hardware Architecture



Reviewed Constraint Analysis

- For **maximum power efficiency**, the model can be operated in a specified timeframe (8 to 10 AM) , during which most indian households have *maximum consumption*.
- The model will operate only for a **maximum water flow pressure of 0.8MPa**.
- The model will operate on an **adjusted temperature range of 6-45℃** which is the usual water conditions.

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Progress Review

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Milestone	Status	Time of Accomplishment
Logical and hardware design phase	✅	Post Phase - I
Core (major) Hardware Connections	✅	Post Phase - I
Hardware Testing and Scoring - Round 1	✅	Post Phase - I
Local hosting server	✅	Post Phase - I
Logging of Arduino data into local system	✅	Post Phase - I
Predictive Analysis of flow data	🕘	Phase = III
Installation of preventive solenoid valve	🕘	Phase = III

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Internal Scoring Scheme

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Developed by Smartopia

Description	Score
Hardware	
Hardware component fails any one test	0
Hardware component passes tests, but without satisfactory results	0.5
Hardware passes all tests satisfactorily.	1
Software	
Software successfully captures and stores **all** serial Arduino data.	1
Software fails to capture **all** serial Arduino data.	0

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Cumulated Scoring

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Attribute	Assigned Score	Comments
Arduino UNO R3	1	Integration with other components is seamless and reliable, low response time.
YF-S201 Flow Sensor	0.5	The sensor detects the change in flow rate, low response time, low accuracy in flow-values.
HW-201 IR Sensor	0.5	Detection range is constricted to 30cm, fast response time.
LEDs - Red and Green	1	The LED lights up at its maximum intensity.
Arduino Buzzer	0.5	The buzzer generates a sharp tone, but relatively low response time.

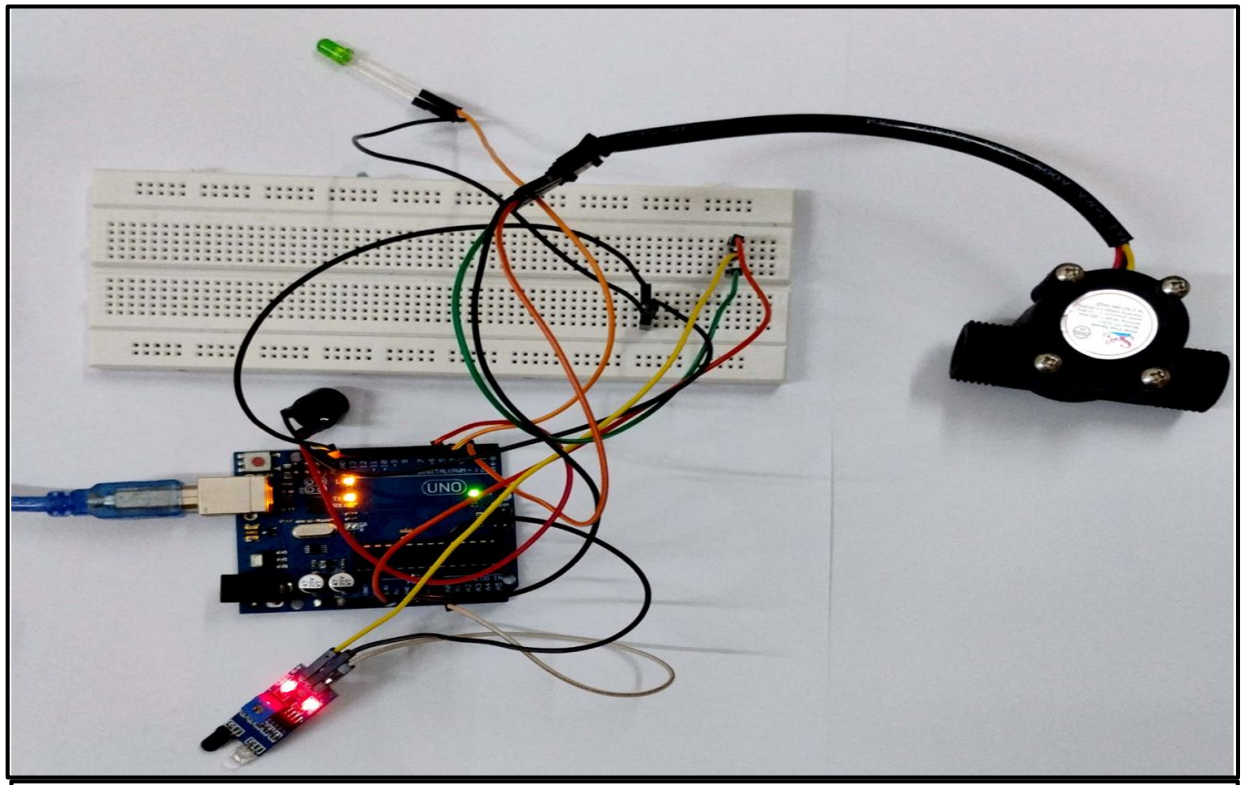
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Pillars of Strength

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Partial Results



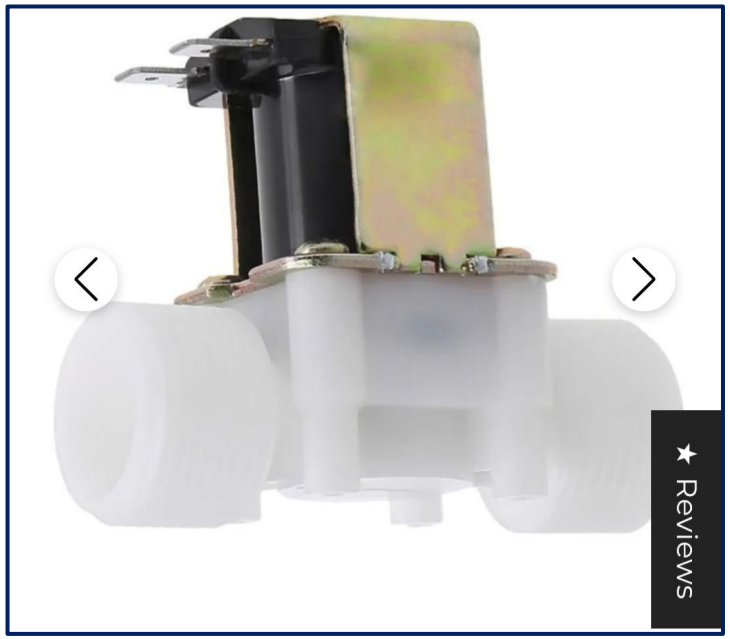
A view of the hardware assembly

Assembled Components
YF-S201
Arduino - UNO R3
Red and Green LED
HW-201

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Imminent Additions

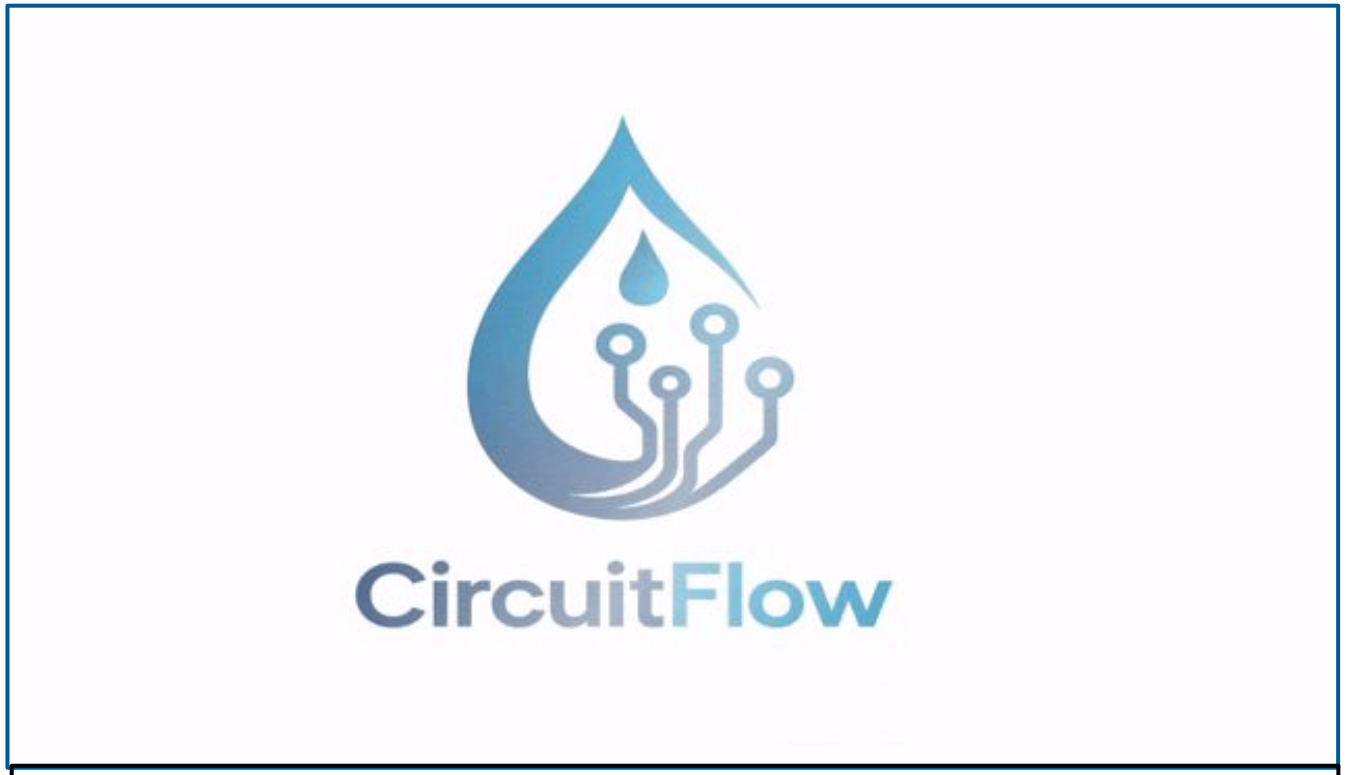
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✅ The Proportional Solenoid Valve

The addition of the valve to the existing hardware shall make the model fully functional upto its designed capabilities.

Partial Results



Extraction of flow-data

Behavioral Signal Extraction

The pyserial library was used for live serial-streaming of Arduino data into the python server program.

Partial Results



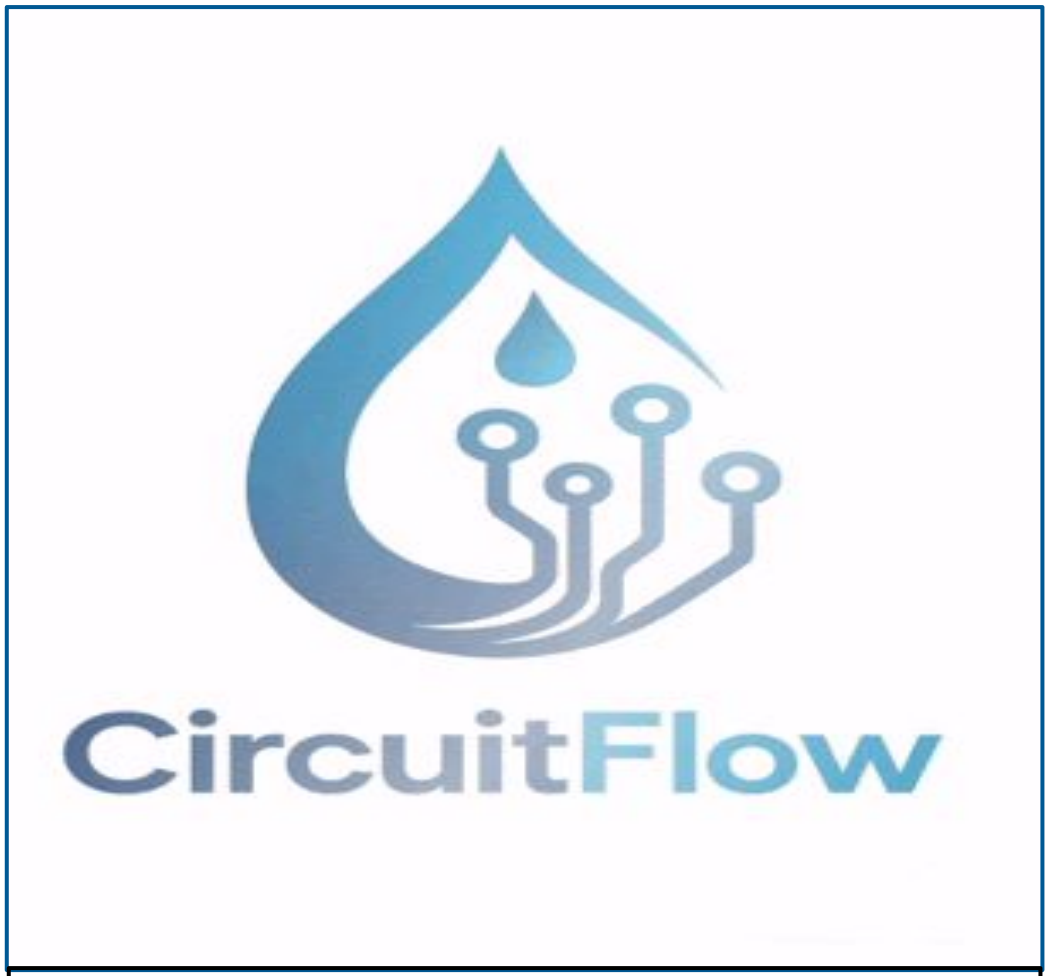
Flow of sensor data from Arduino

Actuation Layer

Sending sensor data from Arduino

The animation depicts the passing of serial data from Arduino, as displayed on the serial monitor, which was achieved by Smartopia.

Partial Results



Interactive User - Interface developed by Smartopia

The Perception Layer

Interactive User - Interface developed by Smartopia

A secure flask server was created to host the basic web pages, which were designed using HTML, CSS and JavaScript.

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Deliverables and Outcomes

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Summary of Key Deliverables of Phase - II
Logical and hardware design
Core (major) Hardware Connections
Hardware Testing and Scoring - Round 1
Local hosting server
Logging of Arduino data into local system

Imminent Work

Imminent Tasks

Completing the hardware circuit *by the injection of the valve.*

Analytical Enhancements: Over the next few days, a predictive model will be developed and tested upon the flow signals received from the Arduino.

Challenges and Considerations

Anticipated challenges in the upcoming phase include ensuring stable and accurate sensor performance at even *low flow rates*, maintaining a reliable *power supply* suitable for continuous operation, and developing *predictive models*

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Objectives: Phase III

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Specific Objectives

- 🕐 Transfer extracted flow data to the presentation layer.
- 🕐 Analytical enhancements towards predictive modeling of data.
- 🕐 Final installation of the protective proportional solenoid valve.

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Conclusion

- ✅**Hardware installation, Data extraction and preliminary software integration:** were successfully completed, setting the stage for real-time data analysis and predictions.
- ✅**Introduction of prefatory server and UI**: Sets the stage for real-time display of flow-data.
- The successful achievements of phase-II pave the way for streamlined flow of work in Phase-III.

References: Phase - I

Authors	Title of the publication	Foundations of the model	Plausible Upgrades/Fixes
Ruturaj Marathe, Manisha Tapale, Vidya Jadhav, Vrushab Hulbatte, Aishwarya Pawar	**IoT based Water Leakage Detection using Smart Objects for Smart City**	The model used Arduino flow-sensors to perceive leakages, and communicated the leakage data to the PushBullet application.	• Integration with solenoid valves to block leaks • analysis of usage-data and • timed water-flow mechanisms.
Chavhan S., Gawande U. H., Bhoyar D., Rangari S., Gardi M., & Gaikwad A.	**IoT-based Remote Control Water Distribution System (**International Journal of Intelligent Systems and Applications in Engineering)	• The model enables the control of water valves remotely through a mobile interface. • The system used a microcontroller(ESP8266), relay drivers, and solenoid valves to control water flow, aiming to reduce water losses and increase response time in saving water in large distribution pipelines.	• It relied on manual control rather than automatic shut off when water is left running. • It is aimed to be handled by the skilled personnel such as water operators, not individuals at the household level. • Integration of daily usage tracking and AI for analysing usage patterns.

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References : Phase - I

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Authors	Title of the publication	Foundations of the model	Plausible Upgrades/Fixes
Vatsala Sharma, Kamal Nayanam, Himani	**Arduino based Smart Water Management**	• The paper lays an excellent foundation for IoT water management with the usage of sensors, though it focuses on automatic water irrigation. • The system uses Arduino (NodeMCU ESP8266), ultrasonic sensors to detect extent of water flow and Blynk app integration.	• Automatic shutoff which reduces wastage. • potential for AI-driven usage optimization • ***local data logging***, making it not entirely reliant on stable internet connection.
Isuru Sachitha Herath	**IoT based Intelligent Domestic Water Management system**	• The paper proposes “Smart Water Buddy”, an IoT and machine learning-based system designed to monitor, analyze, and optimize domestic water usage.	• There is no failsafe mechanism to evade data loss/inconsistencies due to loss of internet connectivity.